



Mapping intertidal topography from optical satellite time series

A per-pixel hypsometric estimator, an openEO pipeline, and why the tide is the limiting error

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Code and data

The problem

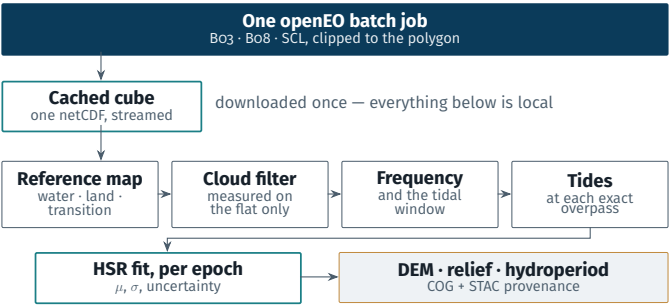
Intertidal flats are among the most productive and most rapidly changing coastal environments, yet they are the hardest to survey. They are too shallow for vessels, too soft and too briefly exposed to walk, and **never wholly visible at any single instant** — the tide hides part of the surface in every image ever taken of it. Their elevation therefore cannot be observed. It has to be *inferred*, by watching how the waterline migrates across many images taken at different tide states, and every method rests on knowing the water level at the moment each image was acquired.

This poster presents an estimator that reads elevation from the full shape of a pixel's wetting curve rather than from a threshold crossing, an open, reproducible pipeline built on openEO, and the finding that the limiting error is now **the assumed water level**, not the estimator.

Why openEO

The archive is the bottleneck: a decade of Sentinel-2 over one estuary is thousands of scenes, of which most are useless. Two common answers both fail at scale — downloading whole scenes wastes almost everything fetched, and closed cloud platforms make the processing itself unciteable. We use **openEO** on the Copernicus Data Space. One batch job requests only the bands and the polygon needed, and returns a single netCDF cube. Everything after that is local:

- Analysis-ready L2A**, subset server-side in space, time and band — no local mirror of the archive.
- Download once, stream forever.** Re-running an analysis never re-downloads; memory is bounded by the time-block size, not by the area or the number of years.
- The cube is the unit of reproducibility**, and the process graph is a portable, standard document rather than a vendor's notebook.



The pipeline. One cloud job, then everything local — which is what makes it affordable to run many contiguous cells in parallel on one machine.

Cloud screening that keeps the data

On an Atlantic coast a scene-level cloud filter is ruinous: discard every image with clouds and little survives. But a cloud over a hillside three kilometres inland does not obscure the flat. So cloudiness is measured **only where it matters**. Two streaming passes:

- every pixel votes across the archive — permanent water, permanent land, or **transition**, which is the intertidal candidate;
- cloud cover is then computed *inside the transition zone alone*, with a dilated safety buffer.

Villaviciosa, nine years, 1379 acquisitions: only 141 are essentially clear over the flat. Demanding clear scenes would leave a decade of imagery yielding almost nothing.

But cloud cover is not a property of a scene — it is a property of each pixel. Masking pixel by pixel inside the transition zone lets partially clouded acquisitions still contribute: **415** scenes have at least half the flat visible, and every clear pixel in them enters the fit. The estimator needs a minimum number of observations *per pixel*, not per scene, which is what makes that possible.

Reproducibility

Every product is a cloud-optimised GeoTIFF with a STAC item beside it carrying the run's resolved parameters, so provenance travels with the file rather than living in a notebook. The toolkit is deliberately granular: no `run_everything()`, one function per scientific step.

Bishop-Taylor, R. et al. (2019). Between the tides: modelling the elevation of Australia's exposed intertidal zone at continental scale. *Est. Coast. Shelf Sci.* 223, 115–128.
Hart-Davis, M. et al. (2021). EOT2o: a global ocean tide model from multi-mission satellite altimetry. *ESSD* 13, 3869–3884.
Schramm, M. et al. (2021). The openEO API. *ISPRS Int. J. Geo-Inf.* 10(3), 125.

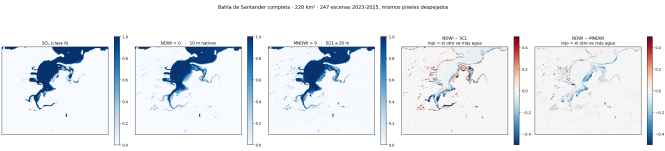
Contains modified Copernicus Sentinel-2 data (2016–2025) via the Copernicus Data Space openEO backend. Tides: EOT2o (SEANOE), GOT4.10 (NASA GSFC) via pyTMD. Field validation: RTK GNSS campaign, Ría de Villaviciosa.

Choosing the water detector

Four detectors were run in **one** backend job over byte-identical dates and cloud masks, so every difference is the index and not the sampling. Scored by rank correlation between water frequency and LiDAR elevation — a near model-free arbiter, since a lower pixel must flood more often:

Index	Spearman ρ	Bands
NDWI	-0.624	Bo3/Bo8, native 10 m
SCL (class 6)	-0.586	scene classification, 20 m
AWEI	-0.584	four-band
MNDWI	-0.559	Bo3/B11, SWIR at 20 m

NDWI wins for a physical reason: both its bands are natively 10 m, so no resampling smears the waterline that carries the signal. MNDWI's SWIR band is the problem — wet sediment reflects like water, so it over-detects. Over the whole Bay of Santander (220 km², 247 scenes) SCL and NDWI agree on 91.7 % of pixels, while MNDWI claims **five times** the intertidal area.



Bay of Santander: the three detectors on identical observations, and the two difference maps. Red is where the other detector sees more water.

HSR: elevation from the whole wetting curve

A 10 m pixel on the waterline is not wet or dry — it is **partially flooded**, and its NDWI is the linear mixture of its wet and dry fractions. As the tide rises that fraction follows the pixel's own internal hypsometry, so its NDWI against tide height is a sigmoid whose parameters are physical, not fitted conveniences:

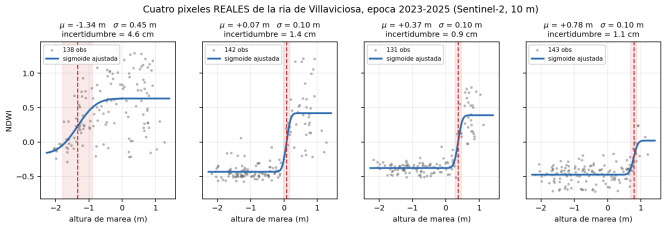
$$\text{NDWI}_p(t) = a_p + b_p \Phi\left(\frac{h_t - \mu_p}{\sigma_p}\right)$$

μ_p — **median pixel elevation**
Read from the *entire* curve rather than from one threshold crossing, so every clear observation contributes and the estimate is sub-decimetre.

σ_p — **sub-pixel relief**
The roughness *inside* the 10 m cell: a product no threshold-crossing method can produce, because it discards exactly the mixed pixels that carry it.

Fit quality — a Cramér–Rao bound per pixel
An uncertainty that is derived rather than assumed, and that doubles as automatic quality control. The amplitude terms (a_p , b_p) have a closed-form least-squares solution for each candidate (μ , σ), so the search is a cheap sweep over a two-dimensional grid instead of an iterative optimiser — which is what makes it tractable over millions of pixels on one machine.

Fitting is done per **epoch**, not per decade: intertidal morphology migrates, and a ten-year fit blurs the transition it is trying to measure. Measured on our data, 87 % of pixels saturate the σ grid at ten years against 12 % at three, and the fit uncertainty halves.



Real per-pixel curves. The sigmoid is not imposed on the data — it is what mixed pixels actually do as the tide crosses them.

Why the LiDAR would not do

The obvious reference for an Asturian estuary is the national 5 m LiDAR DTM. It is **unusable** on the flat: across our intertidal mask it takes only **seven distinct values**, and 83 % of it sits at exactly +2.00 m — not terrain, but the **water surface** at the moment of the flight. Triangulating with water frequency, which must track elevation since a lower pixel floods more often, confirms which reference is the wrong one: $\rho = -0.20$ for the LiDAR against -0.88 to -0.95 for the satellite products. So we went and measured it ourselves.

RTK GNSS field survey

Ría de Villaviciosa, at the lowest spring tide of the month: **361 fixed points at 1.3 cm**. The implied datum offset between survey and satellite is 52.81 m — the local geoid undulation, an independent confirmation that the two describe the same surface in different vertical frames. Each tide model is run through *both* estimators, so every row compares like with like, on the same 107 pixels. RMSE in metres:

Tide model	HSR	Step (DEA)	Δ
GOT4.10	0.159	0.135	0.024 $p = 0.007$
EOT2o	0.135	0.151	-0.016 n.s.
Ensemble	0.138	0.148	-0.010 n.s.

No configuration separates the two estimators under a paired bootstrap, except the one where the baseline benefits from the coarser tide model. Read down the columns instead: HSR moves 0.024 m between tide models and the baseline 0.016 m, **in opposite directions**. The tide model does more than the estimator, and the two do not even respond to it alike.

Of the 361 points, 101 have a water frequency of exactly zero — never once observed wet in nine years. An observational limit of the archive, not a failure of the fit.

The limiting error is the tide

One systematic error is shared by both estimators. Relief comes out **compressed** — the regression of mapped against surveyed elevation has slope 0.7–0.9, not 1 — and the compression worsens ≈ 1.0 of **slope per kilometre** upstream, surviving control for elevation ($t = -3.9$ to -6.6 , $p \leq 0.0002$ across six products). Being identical in both methods, it cannot live in the estimator. It lives in what they share: the assumed water level.

The sigmoid fit is invariant under *affine* maps of the tide axis, which is a theorem rather than a limit of effort — and it says exactly which estuarine effects a satellite can recover:

Estuarine effect	Recoverable?	Measured
Datum offset	No — needs a gauge	—
Constant amplification	No — needs ground truth	—
Phase lag	Yes — flood vs. ebb	≈ 0
Damping with range	Yes — springs vs. neaps	0.25 m

The spring–neap effect alone is twice the RMSE the estimators are separated by. And the published ensemble tide model does not transfer: its ranking dataset is 32 936 points, *all* in Australian waters, so outside that area the weighting collapses and the ensemble becomes an un-weighted mean of its members — difference 0.0000 m over 1379 dates.

The flat as its own tide gauge [in progress]

So stop treating the water level as an input. Let h_t be unknown and solve it jointly with the elevations: which pixels are wet tells you how high the water stood, with tens of thousands voting.



A scene's level is theoretically readable to **5 mm**, against 140 mm for the harmonic model. The obstacle: haze shifts every pixel of a scene together, which to first order is indistinguishable from the water rising. Calibrating on *saturated* pixels, which cannot respond to level, removes it — and agreement with the survey rises from $r = 0.845$ to $r = 0.890$. Not yet conclusive, and reported as such.

Scaling to the north coast [in progress]

Nothing in the method is site-specific and memory is bounded by the time-block size, not by area, so covering a coastline is bookkeeping rather than computation. `aoi.tile()` cuts the coast into contiguous cells, each an independent cube and job; `mosaic.py` then *measures* the datum step across every seam — adjacent tiles each predict the tide at their own centroid — levels it, and averages overlaps instead of double-counting them.

Target: the Cantabrian coast from Ría de Foz to Urdaibai, with Villaviciosa, Santander and Santoña as the first cells. Cell runs and the regional mosaic are in progress.